

Effective depression detection and interpretation: Integrating machine learning, deep learning, language models, and explainable AI

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ABSTRACT

Depression is an increasingly prevalent issue, particularly among young people, significantly impacting their well-being and causing persistent distress. Early detection is crucial to address this growing concern. This study utilizes various machine learning, deep learning, and language models to detect depression among Bangladeshi university students. To address data imbalance in the employed dataset, resampling techniques such as SMOTE and Cluster Centroids are applied. Additionally, exhaustive hyperparameter optimization is performed to enhance classification performance. Our results indicate that machine learning algorithms, particularly Random Forest, effectively predict depression with an accuracy of 91.1% and an F1-score of 91.6%. Language models like RoBERTa also achieve strong results, with a recall score of 98.6%. Moreover, explainable AI (XAI) methods, including SHAP and LIME, are employed to interpret model predictions, underscoring the importance of transparency in machine learning. This work contributes to the early identification of depression by integrating machine learning, deep learning, natural language processing, and XAI techniques. While this study focuses on Bangladeshi or similar demographic groups, the proposed approaches are adaptable and can be applied to other populations for generalization.

1. Introduction

Mental health is a crucial aspect of human well-being. Studies show that those with more life satisfaction also tend to be more engaged at work and have lower rates of worry and despair. On the other hand, people who are less satisfied with their lives are more likely to have poor mental health and lower levels of productivity [1]. Mental health issues, such as depression, can significantly impair a person's ability to think clearly, sleep well, interact with others, and perform daily tasks. Like many other countries, Bangladesh is witnessing a growing number of depression cases, particularly among university students [2–4]. Given the severity of the depression situation in Bangladesh, it is imperative to focus on alleviating depression among Bangladeshi students. Student stress can be triggered by various factors, including socioeconomic, environmental, cultural, psychological, and non-academic characteristics. Stress levels vary significantly depending on students' anxiety symptoms, especially during exam periods. It is reported that 10%–35% of college students suffer from “functionally impairing levels of test anxiety” [5]. Students with high test anxiety are more likely to delay academic progress or drop out of college, which can lead to serious financial consequences and, in some cases, result in

suicidal thoughts or behaviors.

Several studies have focused on the early identification of depression and have proposed various solutions to mitigate its severity. Traditional methods for diagnosing depression often face challenges in terms of time, subjectivity, and accessibility. Artificial intelligence (AI) offers a promising solution by automating and refining these diagnostic processes through advancements in machine learning, deep learning, and natural language processing. These AI-powered systems analyze large datasets to detect patterns associated with depression, providing more accurate and scalable assessments [6,7]. Additionally, personalized AI-driven interventions tailored to individual needs are emerging as a vital trend in mental health support. Machine learning models can now identify depression cases with high accuracy and precision. Socio-demographic factors have been shown to influence the prevalence of depression among students significantly. Therefore, this study utilizes a range of social, demographic, and academic data from Bangladeshi university students to predict depression cases and identify the key contributing factors. In addition to depression prediction using AI methods, it is crucial to conduct thorough data analysis and interpretation of the predictions. This interpretation can be facilitated through

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explainable AI techniques. Ultimately, early detection of depression cases, along with accurate interpretation of the results, can aid in reducing depression among young students.

This study focuses on identifying depression among Bangladeshi university students using a depression dataset that includes information on individuals' physical activities, educational attainment, institutions, monthly incomes, and feelings of depression. Various machine learning algorithms, such as Random Forest and Logistic Regression, are employed to classify whether a student is suffering from depression. In addition to classical machine learning models, deep learning architectures like Multi-layer Perceptron (MLP) and Convolutional Neural Networks (CNN), as well as language models such as BERT, DistilBERT, and RoBERTa, are utilized to predict depression cases. Several evaluation metrics are applied to assess the performance of each model. Explainable AI techniques, including SHAP and LIME, are used to interpret and explain the predictions made by these models. These techniques help identify the impact of each factor contributing to depression. Explainable AI reports provide medical experts and analysts with deeper insights into the factors driving depression and make predictive results more transparent, convincing, and trustworthy.

The major contributions of this work are summarized below:

- An in-depth data analysis was conducted to identify patterns and trends in depression, with various statistical results produced based on the dataset of Bangladeshi university students.
- Several machine learning and deep learning models were used to predict depression among Bangladeshi students based on demographic, social, and academic factors. Hyperparameter optimization and dataset resampling techniques were applied to enhance the models' performance.
- Depression was also predicted using transformer-based language models, utilizing text data instead of tabular data. Consequently, the tabular data was processed into a textual format.
- Two distinct explainable AI techniques, SHAP and LIME, were employed to interpret the prediction results and analyze the contributing factors, making the predictive outcomes easier to understand.

2. Related works

Several studies have focused on predicting depression, particularly among students. A notable contribution by Ku and Min [8] systematically evaluates the performance of various machine learning algorithms in predicting Major Depressive Disorder (MDD) and Generalized Anxiety Disorder (GAD) using self-reported survey data, accounting for subjective response errors. The authors simulate real-world conditions by introducing random perturbations in survey responses to assess the robustness of five distinct predictive models: Convolutional Neural Network, Random Forest, XGBoost, Logistic Regression, and Naive Bayes. Their findings demonstrate that specific hyperparameter adjustments significantly enhance model performance, highlighting the importance of robust machine learning applications in mental health diagnostics.

Vandana et al. [9] proposed a novel approach to depression detection by integrating deep learning techniques, specifically Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, to analyze multimodal data comprising audio and text features. The hybrid model was designed to improve the accuracy of depression classification by utilizing CNNs for feature extraction and LSTMs for sequential data processing. The architecture featured distinct models for textual and audio inputs, which were subsequently combined to form a comprehensive hybrid model. Experimental results demonstrated that the audio CNN model achieved the highest accuracy of 98%. At the same time, the hybrid Bi-LSTM model also exhibited strong performance, underscoring the effectiveness of the proposed method in accurately predicting depression status.

Bhatnagar et al. [10] explored the classification of anxiety levels among university students using machine learning algorithms, emphasizing the significant impact of anxiety on students' daily lives and academic performance. The study applied several machine learning algorithms, including Naïve Bayes, Support Vector Machine, Decision Tree, and Random Forest, with the Random Forest algorithm achieving an accuracy of 78.9%. In another study, Siddiqua et al. [7] addressed the growing concern of depression among students in Bangladesh by utilizing a dataset derived from various established depression assessment scales. They developed a comprehensive questionnaire comprising 106 questions, categorized into personal and clinical inquiries to provide a holistic view of participants' mental health. The study achieved high accuracy, with the Random Forest algorithm reaching 98.08%, demonstrating its effectiveness in predicting depression.

Khan et al. [1] conducted a comprehensive study on life satisfaction prediction using machine learning algorithms and explainable AI methodologies. Utilizing data from a Danish survey, the researchers systematically evaluated several machine learning models to predict binary life satisfaction outcomes. Feature selection techniques, such as Recursive Feature Elimination with Cross-Validation (RFEVCV), along with data resampling methods, were employed to reduce biases and enhance model robustness [11]. An ensemble model achieved strong performance, with an accuracy of 93.80% and a macro F1-score of 73.00%. Additionally, large language models (LLMs) were explored for prediction, with BioBERT achieving a macro F1-score of 73.21%. Explainable AI techniques were incorporated to enhance model interpretability, providing valuable insights into the predictive outcomes.

Joyce et al. [12] extensively reviewed explainable artificial intelligence (XAI) applications. The research found that "explainability" was used in various ways, often with ambiguous definitions. In response, the authors developed the TIFU (Transparency and Interpretability for Understandability) framework, which emphasizes enhancing the interpretability and transparency of AI models to operationalize explainability better. The authors underscore the importance of aligning model presentations with clinical reasoning processes and provide recommendations for implementing the TIFU framework, such as employing transparent models to validate complex AI/ML findings and treating opaque feature representation components as independent preprocessing modules. The study concludes by advocating for greater consistency and specificity in discussions of explainability in mental health AI applications through the TIFU framework.

Banna et al. [13] conducted a comprehensive study on predicting the impact of COVID-19 on mental health. To detect depression in user messages on Twitter, the researchers developed a novel hybrid model that combines Long Short-Term Memory (LSTM) networks with Convolutional Neural Networks (CNN). They systematically analyzed over 571,000 tweets from 482 individuals across four UK locations, spanning from October 2019 to May 2020. The authors integrated two pre-existing datasets and employed various natural language processing techniques for data preprocessing to build their model. The proposed LSTM-CNN model achieved an impressive 99.42% accuracy in classifying tweets as either depressive or non-depressive. By applying their trained model to tweets from both the pre-pandemic and early COVID-19 periods, the researchers observed a significant 18.22% increase in depressive messages and a 55.59% rise in overall social media activity.

Kumar et al. [14] underline the need for early diagnosis and intervention for mental health concerns, notably anxiety and depression. The model for predicting anxious sadness in real-time social data is built by analyzing individuals' social media posting patterns, including strange timing and increased frequency. An Ensemble Vote Classifier combines the outputs of Multinomial Naïve Bayes, Gradient Boosting, and Random Forest. The models demonstrated a fair mix of precision and recall, demonstrating their efficacy in detecting users with anxious sadness. The proposed AD (Anxious Depression) prediction model achieved 85.09% accuracy, with an F1-score of 79.68%.

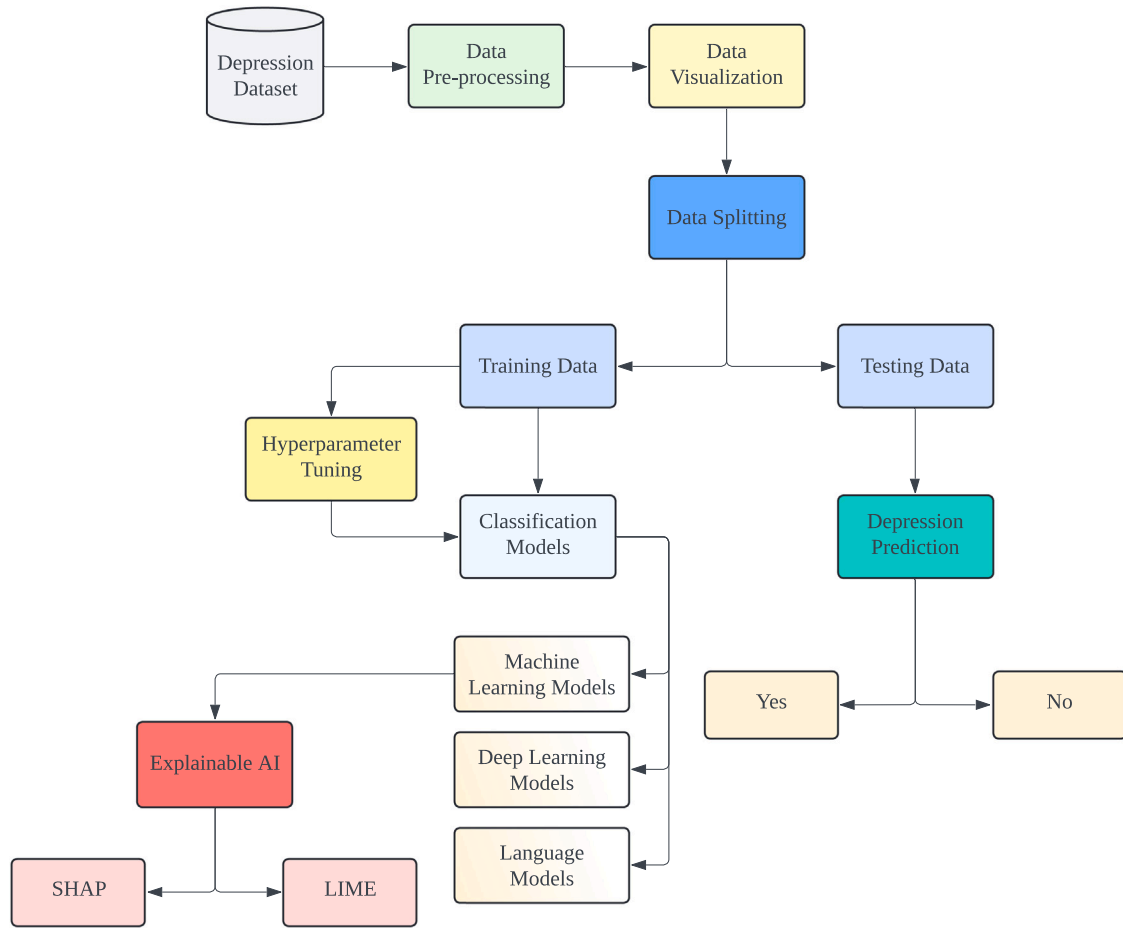


Fig. 1. Methodology flowchart of the study.

Gao et al. [15] reviewed popular machine learning methods for brain imaging classification and predictions, as well as an overview of studies that have used magnetic resonance imaging data to either classify major depression disorder (MDD) from controls or other mood disorders or to investigate treatment outcome predictors for individual patients.

Despite significant advancements in depression prediction tasks, several research gaps persist. First, many studies do not employ explainable AI (XAI) techniques, and those that do typically use only a single method. In our work, we address this by utilizing two distinct XAI techniques, LIME and SHAP, to interpret the predictions made by the machine learning models, thereby enhancing the reliability of the outcomes. Second, much of the current research focuses either on deep learning or classical machine learning techniques in isolation, overlooking the potential benefits of integrating natural language processing (NLP) approaches. This presents a clear opportunity for future studies to successfully combine deep learning, advanced language models, and classical machine learning within a unified framework.

3. Methodology

This study aims to predict depression among Bangladeshi university students by utilizing various machine learning algorithms, deep learning architectures, and computationally intensive language models. The predictions generated by these models are interpreted using two effective explainable AI (XAI) techniques, SHAP and LIME. The complete methodology of this work is illustrated in Fig. 1.

3.1. Dataset

The dataset used in this study is a publicly available dataset and was collected through a comprehensive survey conducted among university students in Bangladesh [16]. It comprises 1602 samples (rows) and 20 distinct variables (columns), capturing various demographic, social, and academic factors. Our study is the first to leverage this specific dataset for machine learning and deep learning-based depression classification, making it a valuable benchmark for future research in mental health analytics. The target variable for this study is the “Feeling Depressed” column, which serves as the label for classification, while the remaining attributes are input features for the machine learning models. The label distribution is imbalanced, with 1149 students reporting “Yes” (feeling depressed) and 453 reporting “No” (not feeling depressed), as depicted in Fig. 2.

The feature variables of this dataset provide a comprehensive representation of the student’s overall well-being, lifestyle, and mental health, making this dataset valuable for research aimed at understanding the factors influencing depression and mental health among university students in Bangladesh. Table 1 presents the participants’ demographic details in the employed dataset. Approximately 52% of the data were collected from male participants, with the remaining 48% from female participants. Most of the data came from students under the age of 25, while only 247 data points were collected from students aged 25 and older. In addition to demographic details, the dataset includes several psychographic factors, such as feelings of loneliness, feelings of anxiety, and the presence of suicidal thoughts. This research will contribute to the growing body of knowledge on mental health,

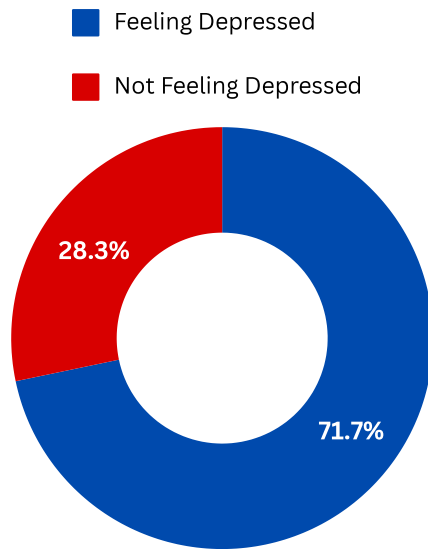


Fig. 2. Percentage of 'depressed' and 'non-depressed' students in the employed dataset.

Table 1
Demographic details of participants.

Attributes	Values	Frequency
Gender	Male	800
	Female	722
Age (years)	22 - 24	890
	< 22	465
	>= 25	247
Education Level	1st/2nd/3rd year	884
	4th/5th/Master's/Higher	718
Current living area	Urban	969
	Rural	633
Monthly family income (taka)	25 to 50 thousand	642
	below 25 thousand	620
	above 50 thousand	340
University type	Private	829
	Public	623
	Others	150

especially in academic settings, and offer insights into the socio-cultural dimensions of depression in a developing country context.

The utilized dataset provides a robust foundation for depression classification using artificial intelligence-based techniques. However, it has certain constraints and limitations. Firstly, the dataset includes data exclusively from Bangladeshi students, which may limit the generalizability of the findings to students from other socio-demographic backgrounds. Secondly, with approximately 1600 data samples, the dataset may not fully capture the diversity of experiences necessary for comprehensive analysis.

3.2. Data pre-processing

The dataset underwent a thorough preprocessing phase to ensure the integrity and reliability of the data for machine learning model development. This initial step involved the removal of both null and duplicate values, which are critical in minimizing data quality issues that can negatively impact model performance. Specifically, 12 null entries and 58 duplicates were identified and removed, resulting in a refined dataset of 1532 entries.

Next, categorical features were transformed into numerical representations using Label Encoder, a common technique used when the data includes categorical variables that need to be represented in a

Table 2

Categorical label encoding for the 'BMI' attribute.

Before Label Encoding	After Label Encoding
Increased	1
Decreased	0
Same as before COVID-19	2

format compatible with machine learning algorithms. This transformation enables the model to interpret these features effectively without introducing biases based on arbitrary numerical values. Table 2 shows the categorical feature transformation for the 'BMI' attribute.

To standardize the feature set, StandardScaler was applied to normalize the data. This scaling technique ensures that features have a mean of zero and a standard deviation of one, which is particularly useful for algorithms that are sensitive to the scale of input data, such as gradient-based models like logistic regression or support vector machines. Scaling the data also helps improve the convergence rate and overall performance of the model during training.

Finally, the processed dataset was split into training and testing subsets using an 8:2 ratio, where 80% (1225 samples) of the data was allocated to the training set and the remaining 20% (307 samples) to the testing set. The training set was utilized to fit (train) the model, while the test set was reserved to evaluate the model's effectiveness and measure its predictive capabilities on new, unseen data.

The entire preprocessing pipeline is essential for ensuring the machine learning model is trained on clean, well-prepared data, which in turn contributes to more reliable and accurate predictions. By addressing potential data quality issues such as missing or redundant values and standardizing feature scales, the model can learn more efficiently and produce more consistent results during evaluation.

Additional Pre-processing for Language Models: For depression prediction using language models like BERT, it is essential to further process the data into a textual format. The original dataset was in tabular form, which was then converted into text to enable the use of language models for the depression prediction task. Each row of the tabular dataset was transformed into a meaningful sentence. Specifically, the dataset contained 20 columns, with the "Feeling Depressed" column used as the target label and the remaining columns serving as features. The feature columns were concatenated to form a coherent sentence. After removing null values and duplicates, a total of 1532 data instances remained, resulting in 1532 sentences. These meaningful textual sentences were used as input features for the language models.

3.3. Data analysis and visualization

Fig. 3(a) presents the proportion of male and female Bangladeshi students in the employed dataset suffering from depression. Notably, 75.5% of female students were found to be experiencing depression, compared to 68.6% of male students. This disparity suggests a higher vulnerability to depression among female students, potentially linked to societal pressures, emotional challenges, or other gender-specific stressors.

Fig. 3(b) examines the prevalence of depression across different stages of education. The data indicates a positive correlation between the level of education and depression rates. Among 1st to 3rd-year students, 70.2% reported experiencing depression, while this figure increased to 73.5% among 4th, 5th-year, and master's level students. This trend may reflect the escalating academic pressure and expectations as students advance in their education, leading to heightened levels of stress and mental health concerns.

Fig. 4 displays the correlation between various attributes and depression, offering insights into key contributing and protective factors. The attribute "feeling anxious" exhibits the strongest positive correlation with depression, with a value of 0.57, suggesting a significant

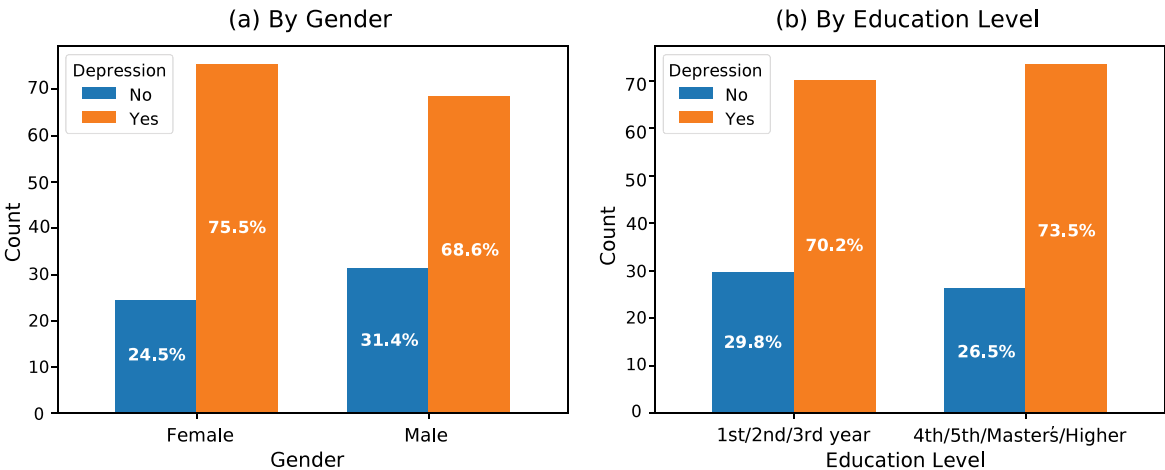


Fig. 3. Percentage distribution of respondents experiencing depression based on (a) Gender and (b) Education Level.

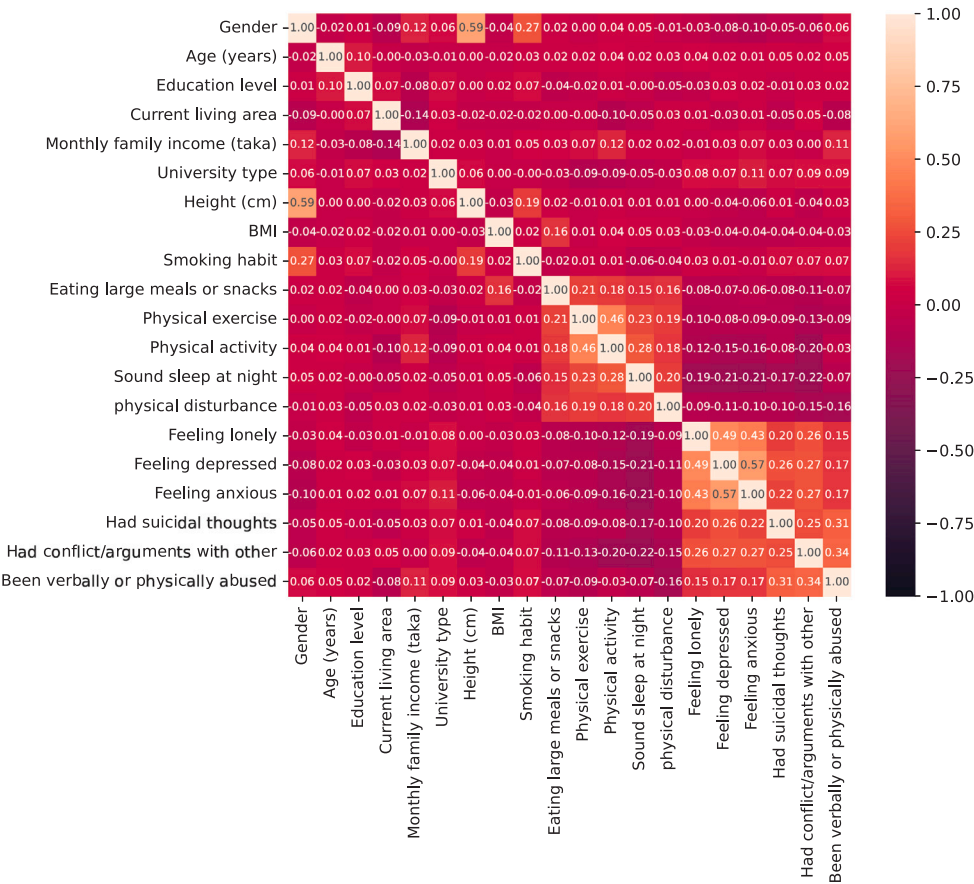


Fig. 4. Correlation matrix showing correlations between all the attributes of the dataset used.

association between anxiety and the likelihood of experiencing depression. Other critical factors include loneliness, with a correlation score of 0.49, conflicts with others (0.27), and suicidal thoughts (0.26), each indicating moderate correlations with depression and highlighting the psychological and social stressors that exacerbate depressive symptoms. Interestingly, smoking habits, while positively correlated with

depression (0.01), show only a minimal association, suggesting that smoking may be more of a consequence than a direct cause of depression. On the other hand, the analysis reveals that sufficient sleep exhibits a negative correlation with depression, with a value of -0.21 , suggesting that students who enjoy restful sleep are less likely to experience depressive symptoms.

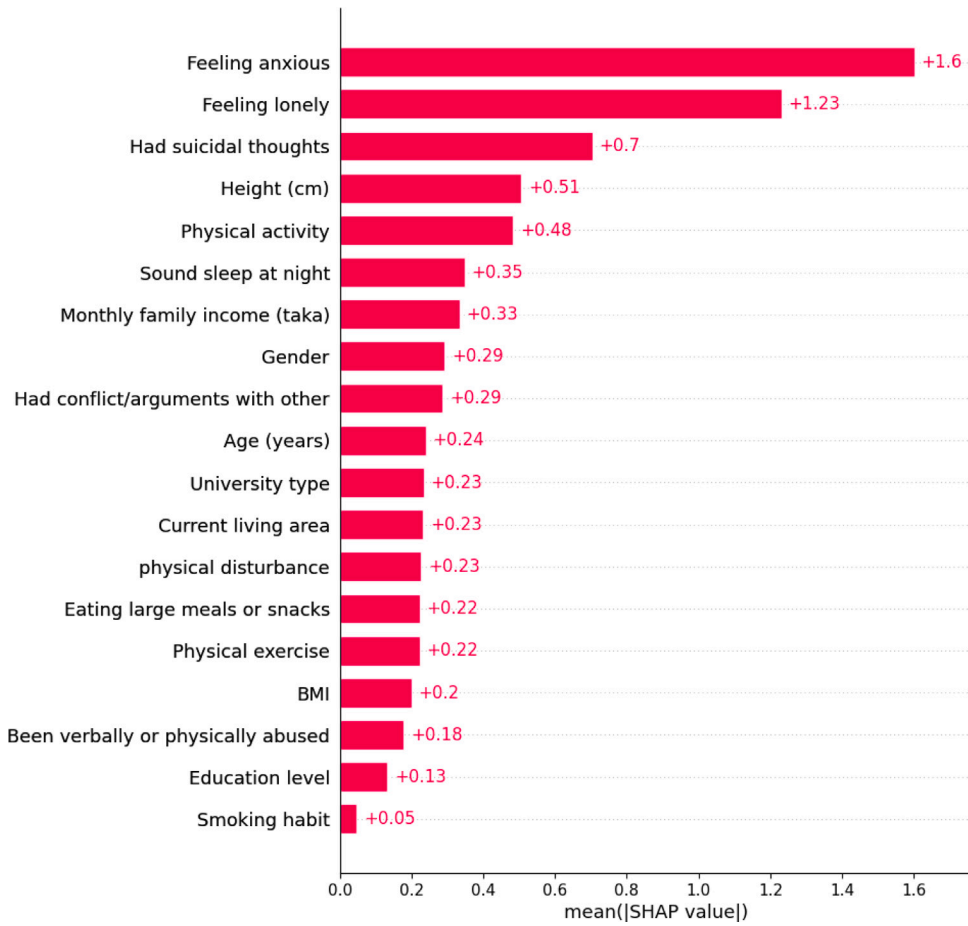


Fig. 5. Feature importance plot generated using SHAP.

Fig. 5 presents the feature importance plot generated using SHAP (SHapley Additive exPlanations) [17], which ranks the impact of various features on depression. The plot orders the features from the highest contributing factor to the lowest, clearly visualizing their relative importance. The top two factors contributing to depression, as indicated by the SHAP values, are feeling anxious (1.6) and feeling lonely (1.23), while the least important factor seems to be the smoking habit (0.05). This analysis aligns with the earlier findings from the correlation matrix presented in Fig. 4, which highlighted anxiety and loneliness as key contributors to depression and smoking habit as a lesser factor.

3.4. Machine learning models

Classical machine learning algorithms were employed to identify depression among Bangladeshi students, serving as a baseline for comparison. For classification tasks on tabular datasets, machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest are often preferred due to their computational efficiency and simplicity. Additionally, Shwartz-Ziv and Armon [18] demonstrated that classical machine learning models, like XGBoost, can sometimes outperform complex deep learning models when applied to tabular data.

Logistic Regression: Logistic Regression [19] is a supervised machine learning algorithm widely used for binary classification problems. It employs the sigmoid activation function to estimate the probability of an outcome. In Logistic Regression, the dependent variable is binary (typically represented as 0 or 1), while the independent variables can be either categorical or continuous. The sigmoid activation function is

mathematically represented in Eq. (1).

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (1)$$

Decision Tree: The Decision Tree [20] is a tree-based supervised machine learning algorithm commonly used for decision-making in both classification and regression tasks. It provides a flowchart-like structure that facilitates decision-making processes, offering a visual and intuitive solution to complex problems. Decision Trees reduce data impurity by recursively splitting the dataset. Two commonly used impurity metrics for this purpose are Gini impurity and Entropy. The mathematical expressions for these metrics are presented in Eqs. (2) and (3).

$$G = 1 - \sum_i p_i^2 \quad (2)$$

$$E = \sum_i -p_i * \log_2 p_i \quad (3)$$

Random Forest: Random Forest [21] is an ensemble learning method that makes decisions based on a collection of decision trees. It is a widely used algorithm for both classification and regression tasks. The method leverages two key concepts: Bagging (bootstrap aggregation) and random feature selection. Random Forests outperform individual decision trees by reducing overfitting, as they combine predictions from multiple trees, each trained on slightly different subsets of the data and features.

XGBoost: XGBoost (eXtreme Gradient Boosting) [22] is an optimized, distributed gradient boosting library designed for efficient and scalable machine learning. It implements machine learning techniques

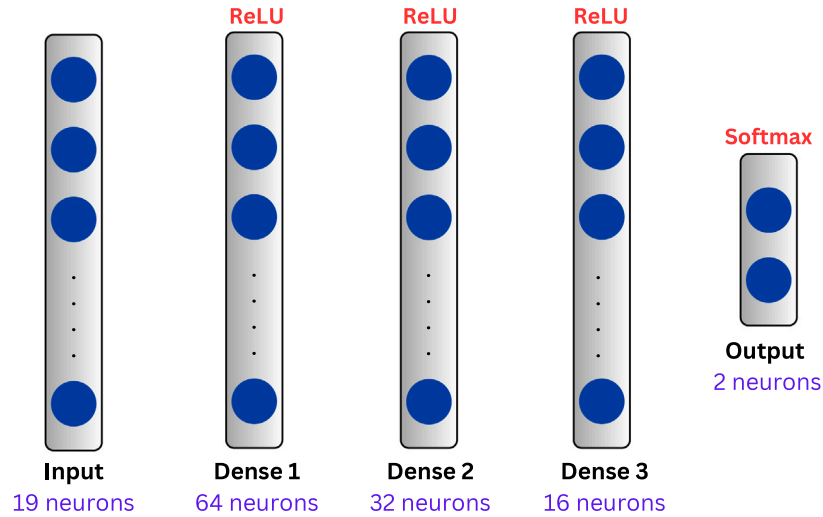


Fig. 6. Architecture of the custom MLP model.

within the Gradient Boosting framework. XGBoost distinguishes itself from traditional gradient boosting methods through several advanced features, including sophisticated tree regularization and enhanced randomization settings. These characteristics contribute to its superior performance in both classification and regression tasks.

AdaBoost: AdaBoost (Adaptive Boosting) [23] is a machine learning meta-algorithm used for classification. It combines multiple “weak” classifiers to form a more robust, accurate classifier. Originally designed for binary classification, AdaBoost has been extended to handle multi-class classification and regression tasks. The algorithm iteratively adjusts the weights of each weak learner based on its performance, with subsequent rounds focusing more on the misclassified instances. This adaptive process enables AdaBoost to produce a final model that is generally more accurate than its individual components.

K-Nearest Neighbor: The K-Nearest Neighbors (KNN) [24] technique is a non-parametric machine learning method used for both classification and regression tasks. In KNN, an object is classified based on a majority vote of its neighbors, assigning it to the class with the highest frequency among its k nearest neighbors. For regression tasks, the prediction is the average of the values of the k closest neighbors. KNN is categorized as an instance-based learning method, with computations deferred until the classification stage, making it a lazy learning algorithm.

Support Vector Machine: Support Vector Machines (SVMs) [25] are supervised learning models used for both classification and regression analysis. SVMs operate by identifying a hyperplane that optimally separates different classes in a high-dimensional space. They are particularly effective in high-dimensional settings, significantly when the number of dimensions exceeds the number of samples [26]. SVMs can also perform non-linear classification efficiently using the kernel trick, which implicitly maps inputs into higher-dimensional feature spaces. This capability makes SVMs well-suited for both linear and non-linear classification tasks.

3.5. Deep learning models

Deep learning models, specifically Multi-Layer Perceptron (MLP) and Convolutional Neural Networks (CNN), were employed for the depression detection task due to their strong ability to capture complex patterns in datasets. In scenarios where classical machine learning models may fall short, deep learning models often deliver more

robust performance. Comparing the performance of MLP and CNN with classical machine learning models allows for a deeper understanding of which algorithms are best suited for depression detection tasks.

Multi-layer Perceptron (MLP): A feedforward neural network with multiple layers, typically consisting of an input layer, one or more hidden layers, and an output layer, is known as a Multi-layer Perceptron (MLP) [27]. MLPs extend single-layer perceptrons to handle non-linearly separable data. These networks are widely used for various tasks, including image recognition and time-series analysis, and can approximate complex functions. In our experiment, we designed a custom MLP model to classify depression among Bangladeshi university students. The architecture has a total of 3922 trainable parameters. The proposed MLP consists of an input layer, an output layer, and three dense hidden layers. The hidden layers utilize the ReLU activation function, while the output layer employs the Softmax activation function, as shown in Fig. 6.

Convolutional Neural Network (CNN): A Convolutional Neural Network (CNN) [28] is a widely utilized deep learning architecture, primarily designed to process grid-like inputs such as images. CNNs leverage convolutional techniques to extract hierarchical information, drawing inspiration from the visual cortex. In a typical CNN design, convolutional layers are responsible for feature detection, pooling layers reduce dimensionality, and fully connected layers are used for higher-level reasoning. CNNs have demonstrated exceptional performance in computer vision tasks such as image segmentation, object detection, and image classification. Our proposed CNN architecture consists of 18,818 trainable parameters, featuring two convolutional layers followed by two MaxPooling layers, which are then connected to a fully connected neural network. The CNN architecture employed in this study is depicted in Fig. 7.

3.6. Language models

Language models were employed to identify depression among Bangladeshi students, offering a unique approach to this task. Unlike machine learning and deep learning models, which primarily operate on tabular datasets, language models are designed to work with textual data. To leverage this capability, the tabular dataset was transformed into meaningful textual representations, enabling its use with language models. While language models are less computationally efficient, they were included in this study to explore how performance varies when



Fig. 7. Architecture of the custom CNN model.

textual data is used instead of tabular data. This approach is unique for depression classification tasks and is absent in most previous works.

BERT: Bidirectional Encoder Representations from Transformers (BERT) [29] is a widely used language model, particularly effective for natural language understanding (NLU) tasks [30] such as sentiment classification, natural language inference, and named entity recognition. Given BERT's outstanding performance in various classification tasks, it is a logical choice for classifying depression among students. In our study, we employed pre-trained BERT models, which were trained using masked language modeling (MLM) and next-sentence prediction. The base version of BERT used in this study consists of 110 million parameters.

DistilBERT: DistilBERT [31] is a compressed and efficient version of the BERT model, designed to reduce computational requirements without significantly sacrificing performance. Approximately 40% smaller than the base BERT, DistilBERT retains around 97% of BERT's language processing capabilities, having been developed using knowledge distillation techniques [32]. During pre-training, the DistilBERT model was guided by both ground truth data and the teacher model (BERT). This approach enables DistilBERT to leverage the inductive biases learned by larger models, while simultaneously reducing resource usage and speeding up inference times. With 67 million parameters, DistilBERT is notably more computationally efficient than BERT, making it an alternative for tasks requiring reduced processing power.

RoBERTa: The Robustly Optimized BERT Approach (RoBERTa) [33], developed by Facebook AI, is an advanced model based on the BERT architecture. The most notable distinction between BERT and RoBERTa is that RoBERTa is pre-trained on a significantly larger corpus than BERT. It was pre-trained using the masked language modeling (MLM) technique and has demonstrated state-of-the-art performance across various NLP tasks. The RoBERTa base model used in this study consists of 125 million parameters.

3.7. Imbalanced dataset resampling

In our study, we observed a significant class imbalance in the dataset, with the target variable showing a distribution of 71.6% 'Yes' responses and 28.4% 'No' responses. To address this data imbalance and minimize its potential effect on model performance, we applied two resampling techniques: oversampling and undersampling.

Data oversampling is a widely utilized technique in machine learning for addressing class imbalances, particularly when one class is significantly underrepresented. In this study, we employed the Synthetic Minority Over-sampling Technique (SMOTE) to address the imbalance in our dataset. SMOTE generates synthetic data points for the minority class, enhancing the representation of the minority class without replicating existing observations. This approach often leads to improved model performance by allowing the model to capture patterns in the minority class better.

In addition to oversampling, we implemented undersampling strategies to assess the effectiveness of various resampling techniques on the label distribution. Among the available methods, we employed the Cluster Centroids approach [34], which clusters the majority class and defines each centroid as the arithmetic mean of the data points within the cluster [35]. This technique offers notable advantages, including ease of implementation and computational efficiency, particularly for large datasets. Compared to other undersampling techniques, the Cluster Centroids method has demonstrated superior performance in several cases, making it a promising choice for handling class imbalances. Fig. 8 illustrates the class distribution for the "Feeling Depressed" variable, both before and after applying resampling techniques.

Constraints of Resampling Approaches: While SMOTE and Cluster Centroids are effective in mitigating class imbalance, both techniques come with inherent constraints and limitations. SMOTE generates synthetic data points for the minority class to create a more balanced dataset. However, these synthetic samples may not always accurately represent real-world data, potentially introducing biases or noise into the dataset. On the other hand, Cluster Centroids addresses class imbalance by undersampling the majority class, which involves removing certain data points. This approach risks discarding valuable information from the majority class. Furthermore, the reduction in dataset size can lead to a less diverse dataset, which is a critical limitation for studies in human mental health that require diverse data patterns to capture complex behaviors and characteristics. The resulting smaller dataset can also negatively impact performance metrics, such as accuracy and F1 scores, as observed in our experiments.

3.8. Hyperparameter optimization

Hyperparameter tuning [36] is a crucial step in maximizing machine learning models' performance. Achieving high accuracy and generalization on unseen data requires selecting the optimal set of hyperparameters, such as regularization strength, kernel type, and learning rates. Machine learning models typically perform better after hyperparameter optimization.

In this work, an extensive hyperparameter optimization was conducted using GridSearchCV [37] for all the machine learning models to obtain the best results. GridSearchCV searches for the optimal hyperparameter values by constructing a grid of all possible combinations. The best hyperparameter combination is selected based on the evaluation results obtained through cross-validation. Generally, applying the optimal set of hyperparameters to a machine learning model improves its performance.

Although other hyperparameter tuning techniques like RandomizedSearchCV exist, we employed GridSearchCV because it thoroughly searches all possible combinations of hyperparameters, despite being computationally expensive. Our experiments showed significant improvements in model performance after optimizing the hyperparameters. The explored hyperparameter space and the optimal set of hyperparameters used for all machine learning models are summarized

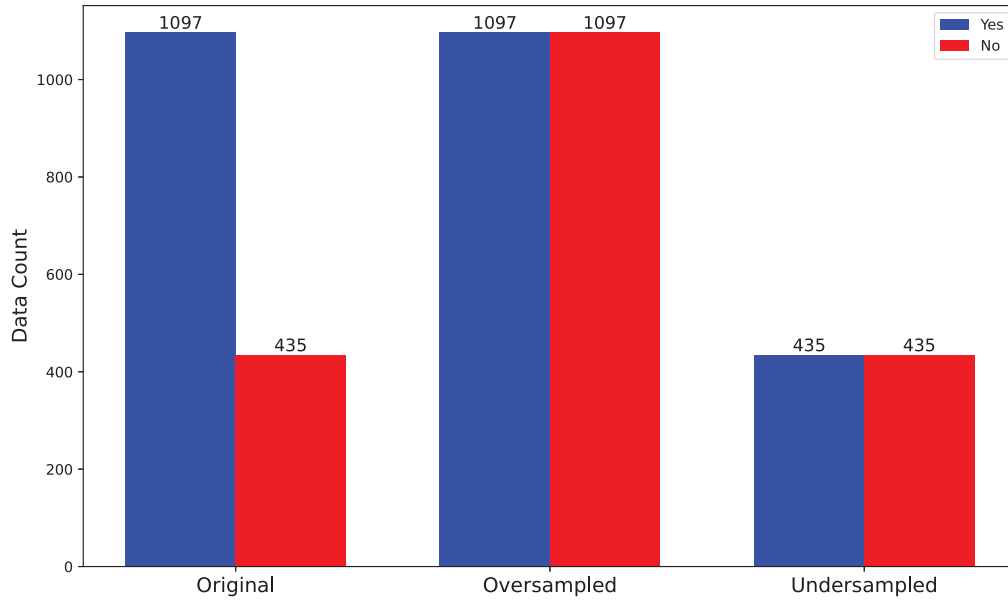


Fig. 8. Bar plot for the class distribution of 'Yes' and 'No' data counts in the original, oversampled, and undersampled datasets.

Table 3

Exploration of the hyperparameter space and the optimal hyperparameters for all employed machine learning models.

Model	Hyperparameter Space	Best Hyperparameter
KNN	n-neighbors: 1–12	n-neighbors: 1
SVM	kernel: linear, poly, rbf, sigmoid gamma: scale, auto	kernel: rbf gamma: scale
AdaBoost	n-estimators: 10, 50, 100, 200 learning-rate: 0.001, 0.01, 0.1, 1	n-estimators: 100 learning-rate: 1
XGBoost	n-estimators: 10, 50, 100, 200 learning-rate: 0.001, 0.01, 0.1, 1	n-estimators: 50 learning-rate: 0.1
Decision Tree	criterion: gini, entropy max-depth: 1–12	criterion: gini max-depth: 5
Random Forest	criterion: gini, entropy max-depth: 1–12 n-estimators: 10, 50, 100, 200	criterion: gini max-depth: 11 n-estimators: 100
Logistic Regression	solver: lbfgs, newton-cg, liblinear, sag, saga	solver: liblinear

in Table 3. We used the default values for the other hyperparameters not listed in the table. The range of hyperparameters was selected based on prior studies in this domain to ensure relevance and effectiveness. A consistent range was applied across models to maintain fairness during the optimization process.

4. Experiment

This section provides a detailed overview of our proposed experiments, including comprehensive evaluation results and their analysis.

4.1. Implementation details

In this work, we predicted depression using two deep learning model architectures: Multi-layer Perceptron (MLP) and Convolutional Neural Network (CNN). All experiments were conducted using the *PyTorch* framework. The models were trained on a training set containing 80% of the data and evaluated on a test set containing the remaining 20%. Both the MLP and CNN models used in this study were lightweight, allowing for training and evaluation without needing a dedicated GPU. Table 4 presents the hyperparameters used for implementing the deep learning models.

In addition to machine learning and deep learning models, we employed language models to predict depression based on textual data, employing three popular models: BERT, DistilBERT, and RoBERTa. These models were fine-tuned on the training set using the HuggingFace Trainer API, with the code implemented in the *PyTorch* framework. We used cross-entropy loss and the AdamW optimizer, with a learning rate of $5e-5$, based on prior studies that employed language models for classification tasks. Training was conducted for 10 epochs, as performance stopped improving beyond that point. A batch size of 8 was selected due to computational constraints, as our system could not accommodate larger batch sizes. Table 4 presents the hyperparameters used for implementing the language models. Since these language models contain a significant number of parameters and are computationally intensive, they require a dedicated GPU. We used an NVIDIA RTX 3080 GPU with 12 GB of VRAM for our experiments.

4.2. Results and analysis

This section provides an in-depth analysis of the results for classifying depression among Bangladeshi university students. We compare the classification performance of seven distinct machine learning models, two deep learning models, and three language models.

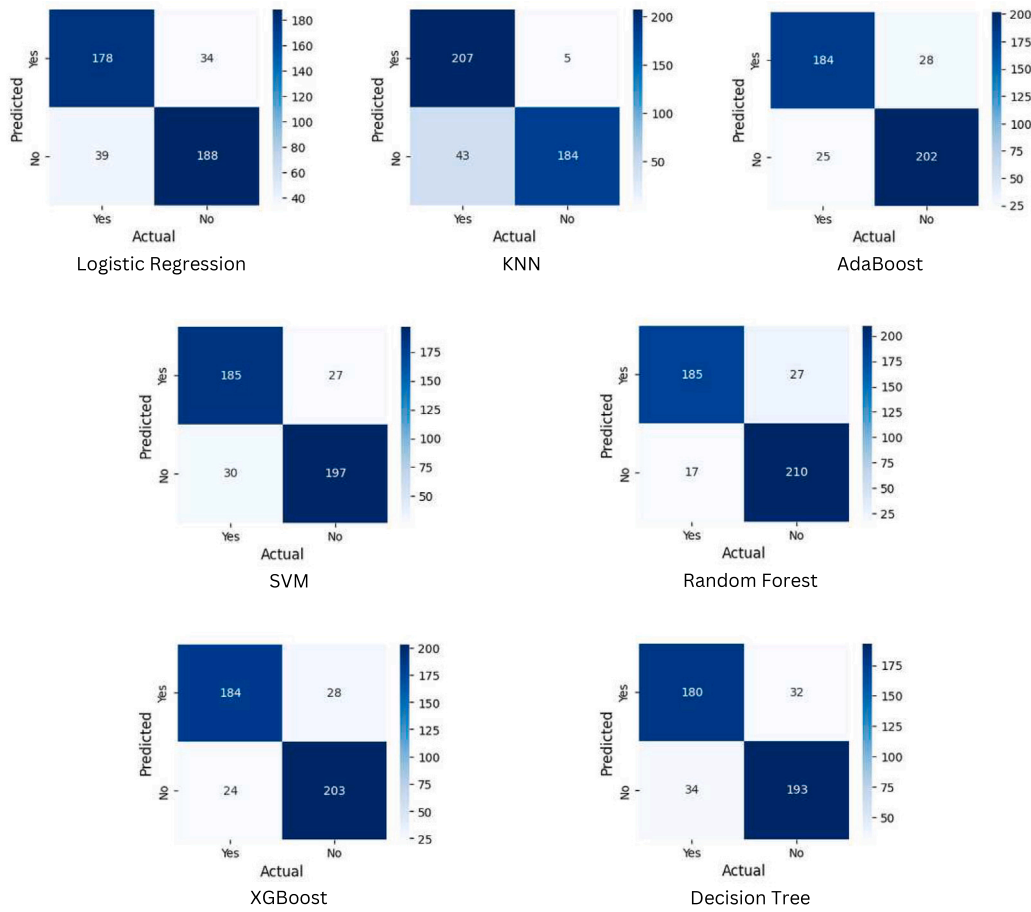


Fig. 9. Confusion matrices of the applied machine learning models.

Table 4

Hyperparameters used for the employed deep learning and language models.

Hyperparameter	Deep Learning Models Value	Language Models Value
Loss	Cross-Entropy	Cross-Entropy
Epochs	40	10
Batch Size	32	8
Optimizer	Adam	AdamW
Learning Rate	0.001	5e-5

4.2.1. Performance of machine learning models

The results achieved by the machine learning models are summarized in Table 5. Five key evaluation metrics — accuracy, precision, recall, F1 score, and AUROC — were used to assess model performance. As shown in Table 5, Random Forest outperforms the other machine learning models in terms of accuracy, F1 score, recall, and AUROC, while K-Nearest Neighbor (KNN) achieves the highest precision of 97.4%. Overall, it can be concluded that among the classical machine learning algorithms used, Random Forest demonstrates the best performance, with an accuracy of 91.1% and an F1 score of 91.6%, highlighting its robustness and suitability for predicting depression-related outcomes. The performance variability of the best-performing Random Forest model across the 95% confidence interval is presented in Table 6. Fig. 9 presents the confusion matrices for the machine learning models employed.

4.2.2. Performance of deep learning models

In our experiments, we evaluated the performance of two deep learning models: the Multi-layer Perceptron (MLP) and the Convolutional Neural Network (CNN). As presented in Table 5, the MLP

achieved an accuracy of 81.8%, while the CNN reached 83.1%. Despite these models' effectiveness in domains like image classification, their performance in tabular dataset classification tasks is notably lower than that of the best-performing machine learning model. This result suggests that while deep learning models can be powerful in certain contexts, they may fall short in handling structured data, where machine learning approaches might provide better results.

4.2.3. Performance of language models

For our depression classification task, we employed three distinct language models: DistilBERT, BERT, and RoBERTa, as illustrated in Table 7. Among these, DistilBERT is the most computationally efficient with the lowest number of parameters (67 million). Although it is particularly suitable for memory-constrained environments, it does not outperform the other language models, BERT and RoBERTa.

According to Table 7, BERT achieved the highest accuracy at 85.3%, along with an impressive F1 Score of 90.6% and a precision of 87.9%, highlighting its effectiveness for classification tasks. While BERT excelled in three evaluation metrics, RoBERTa achieved the highest recall value among the three models at 98.6%, demonstrating its strength in identifying true positives and minimizing false negatives. Overall, BERT stands out for its balanced performance in accuracy and F1 Score, while RoBERTa's high recall makes it particularly valuable in scenarios where identifying all relevant instances is crucial. DistilBERT, while offering a more computationally efficient alternative, delivered reasonable performance but did not match the results of BERT and RoBERTa. The language models generally exhibit higher recall values than the machine and deep learning models.

After contrasting the evaluation results presented in Tables 5 and 7, it is evident that the Random Forest model overall performs better than

Table 5

Classification results of the machine learning and deep learning models on the test-set of the employed dataset.

Model	Accuracy (%)	F1 (%)	Precision (%)	Recall (%)	AUROC
KNN	89.1	88.5	97.4	81.1	0.89
SVM	87.0	87.4	87.9	86.8	0.87
AdaBoost	87.9	88.4	87.8	89.0	0.88
XGBoost	88.2	88.6	87.9	89.4	0.88
Decision Tree	85.0	85.4	85.8	85.0	0.85
Random Forest	91.1	91.6	89.5	93.8	0.90
Logistic Regression	83.4	83.7	84.7	82.8	0.83
MLP	81.8	87.8	87.1	88.6	0.75
CNN	83.1	88.8	87.3	90.4	0.76

Table 6

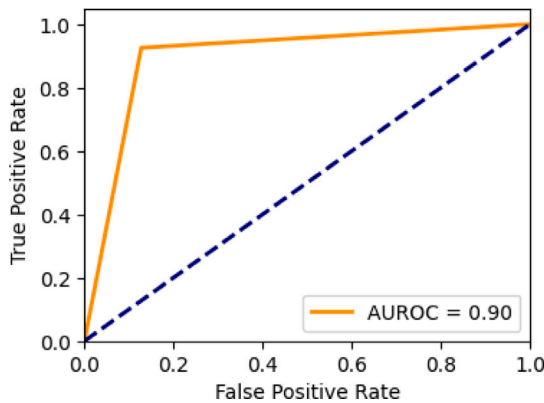
Random Forest model performance variability across 95% confidence intervals.

Evaluation Metric	Value (%)	95% Confidence Interval (%)
Accuracy	91.1	88.8–94.2
F1 Score	91.6	89.5–94.6
Precision	89.5	88.1–93.8
Recall	93.8	90.5–96.8

Table 7

Classification results of the language models on the test-set of the employed dataset.

Model	Param (M)	Accuracy (%)	F1 (%)	Precision (%)	Recall (%)
DistilBERT	67	82.1	87.5	83.5	91.9
BERT	110	85.3	90.6	87.9	93.5
RoBERTa	125	84.7	90.1	83.0	98.6

**Fig. 10.** ROC curve for the best-performing 'Random Forest' model.

all other models. The receiver operating characteristic (ROC) curve for Random Forest, shown in Fig. 10, indicates an area under the curve (AUC) of 0.90.

While direct comparisons with existing studies are challenging due to differences in datasets and input features, prior research on depression detection has shown that performance varies significantly based on data characteristics and feature selection techniques. Our model's accuracy of 91.1% highlight the effectiveness of our data resampling and hyperparameter optimization strategies. These results are consistent with or surpass trends observed in related studies, demonstrating the effectiveness of our approach.

Trade-offs Between Evaluation Metrics: In the context of depression prediction, false negatives are more critical than false positives, as failing to identify individuals with depression can have serious future consequences. Therefore, recall can be considered more crucial than precision in this domain. While lower precision may lead to over-predicting positive cases — which can be undesirable in certain applications — high recall ensures that most cases of depression are identified, minimizing the risk of undiagnosed individuals. Among the three language models used, RoBERTa demonstrated the highest recall

of 98.6%, alongside a precision of 83%, making it particularly effective in accurately identifying depression cases. Furthermore, its F1-score of 90.1% reflects a balance between precision and recall, ensuring both comprehensive and reliable detection. However, while high recall is beneficial for capturing most cases, it is essential to balance it with precision to avoid unnecessary misclassifications, especially in sensitive mental health applications.

4.3. Impact of data resampling techniques

In this study, we addressed the class imbalance in our dataset using two resampling strategies: SMOTE for oversampling and Cluster Centroids for undersampling. This section provides a comparative analysis of how each resampling approach affected the overall performance of the machine learning models. Table 8 summarizes the models' performance on the original, oversampled, and undersampled datasets.

As shown in Table 8, undersampling with Cluster Centroids did not yield satisfactory results in our experiments. It led to lower accuracy, F1 score, and AUROC across all employed models, except for AdaBoost, compared to the results achieved with the original dataset. This suggests that AdaBoost can operate effectively even with smaller datasets. On the other hand, KNN experienced the most significant decline, with accuracy dropping by 10.3%. The overall poor performance observed with the undersampled data can be attributed to the small size of the original dataset, which was further reduced through undersampling, thereby limiting the models' ability to generalize effectively. In contrast, the SMOTE oversampling strategy consistently outperformed both the original and undersampled datasets across all evaluation metrics—accuracy, F1 score, and AUROC. Among the models trained on the oversampled dataset, Random Forest achieved the best performance, with an accuracy of 91.1% and an F1 score of 91.6%. These results indicate that machine learning algorithms, especially tree-based models, benefit significantly from training on larger, balanced datasets.

While oversampling with SMOTE effectively addresses data imbalance and enhances model performance, it also increases computational complexity. Conversely, undersampling with Cluster Centroids reduces training time but results in suboptimal performance. Based on our comparative analysis, we conclude that SMOTE-based oversampling is the most effective resampling approach for our experiments, despite its higher computational cost. The observed improvements in accuracy, F1 score, and AUROC across all models underscore the importance of addressing class imbalance to achieve optimal results.

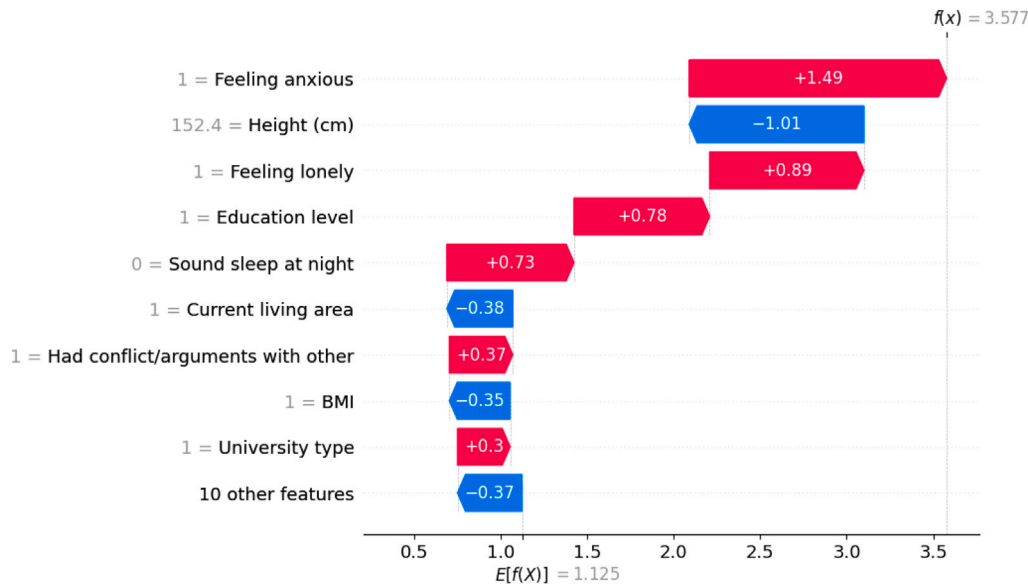
4.4. Explainable AI

Until now, our model made predictions and decisions without clearly interpreting how the results were generated. We applied Explainable AI (XAI) techniques to understand our model's decision-making process better and obtain more reliable results. This approach not only improved the interpretability of our model but also provided more profound insights into the factors influencing its predictions. Our research utilized two distinct explainable AI techniques: SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations).

Table 8

Performance comparison of machine learning models after applying oversampling and undersampling techniques to address data imbalance.

ML Model	Original			Oversampling			Undersampling		
	Acc. (%)	F1 (%)	AUROC	Acc. (%)	F1 (%)	AUROC	Acc. (%)	F1 (%)	AUROC
KNN	82.1	88.0	0.76	89.1	88.5	0.89	71.8	72.0	0.72
SVM	82.1	88.0	0.76	87.0	87.4	0.87	74.1	74.0	0.74
AdaBoost	83.7	89.1	0.78	87.9	88.4	0.88	83.9	83.1	0.84
XGBoost	82.1	88.4	0.73	88.2	88.6	0.88	79.9	79.0	0.80
Decision Tree	80.8	86.6	0.78	85.0	85.4	0.85	74.1	73.1	0.74
Random Forest	84.0	89.7	0.75	91.1	91.6	0.90	77.6	76.4	0.78
Logistic Regression	81.4	87.5	0.76	83.4	83.7	0.83	74.1	73.7	0.74

**Fig. 11.** SHAP interpretation of a positive depression prediction case.

4.4.1. SHAP analysis

SHAP is a widely used explainable AI framework that interprets any machine learning model by assigning contribution value to each feature in relation to the prediction, enabling fair attribution [17], [38]. Figs. 11 and 12 present the SHAP interpretations of depression predictions for a positive case (suffering from depression) and a negative case (not suffering from depression), respectively. In the positive case shown in Fig. 11, anxiety, loneliness, and education level were the top three factors contributing to the prediction of “Yes”, while the person’s height contributed to a “No” prediction. However, due to the more substantial overall influence of the positive factors, the model predicted that the person was suffering from depression. Conversely, Fig. 12 shows the interpretation for a negative case. In this instance, the individual’s anxiety status, loneliness status, and family’s monthly income were the most vital contributors to the prediction of “No”. As a result, the model predicted that this person was not suffering from depression.

Figs. 13 and 14 present another valuable interpretation using the SHAP beeswarm plot. The beeswarm plots show the top 5 impactful attributes impacting a student’s depression status. From the plots, it is evident that ‘anxiety’ has the highest impact on a student’s depression status, while ‘loneliness’ is the second most impactful feature.

4.4.2. LIME analysis

LIME is an easy-to-understand, explainable AI framework that explains individual predictions by approximating a complex model with a simpler, interpretable one [39]. This technique has been beneficial for understanding how the model behaves for specific data points. LIME interpretations for a positive depression case and a negative depression case are presented in Figs. 15 and 16, respectively. In Fig. 15, the machine learning model classifies the case as positive (Yes) due to

the strong influence of factors such as anxiety, suicidal thoughts, and loneliness. Notably, the top 8 significant factors contributed to the positive class, leading to the model’s prediction. On the other hand, Fig. 16 shows a case where the machine learning model predicts a negative class (No). The model made this prediction because the person was not experiencing anxiety, suicidal thoughts, or loneliness, resulting in a negative class prediction.

These XAI techniques, such as LIME or SHAP, not only make the model more interpretable but also provide valuable insights for counselors and healthcare providers. By understanding the factors behind a model’s predictions, mental health professionals can better assess depression, offer more personalized care, and build trust in AI-driven systems. In the end, these approaches help ensure more informed, reliable, and compassionate decision-making in mental health care.

5. Limitations

While our study demonstrates strong predictive performance in detecting depression using machine learning, deep learning, and language models, several limitations must be acknowledged. (1) Although our model demonstrated strong predictive performance, its generalizability to different populations remains unevaluated due to dataset limitations. Future studies could explore the model’s performance on datasets representing different demographics to evaluate its reliability. (2) Furthermore, the dataset may contain demographic biases that could impact the model’s predictions. Addressing this bias through bias mitigation techniques is an important direction for future research. (3) Fine-tuning large-scale language models such as RoBERTa requires significant computational resources, which can be a significant limitation for real-world deployment. (4) Moreover, there is a need to investigate

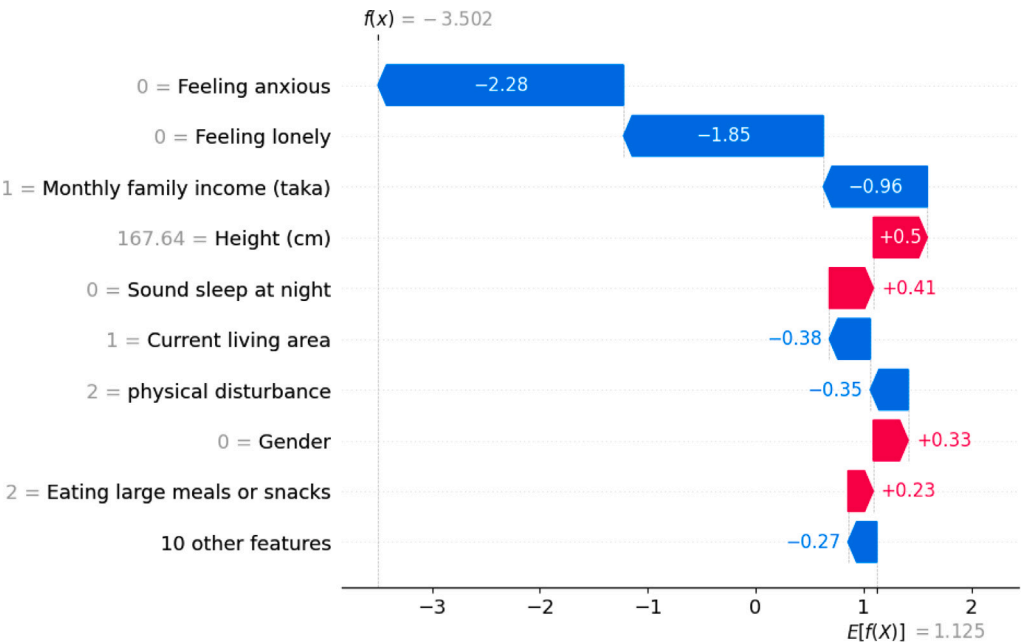


Fig. 12. SHAP interpretation of a negative depression prediction case.

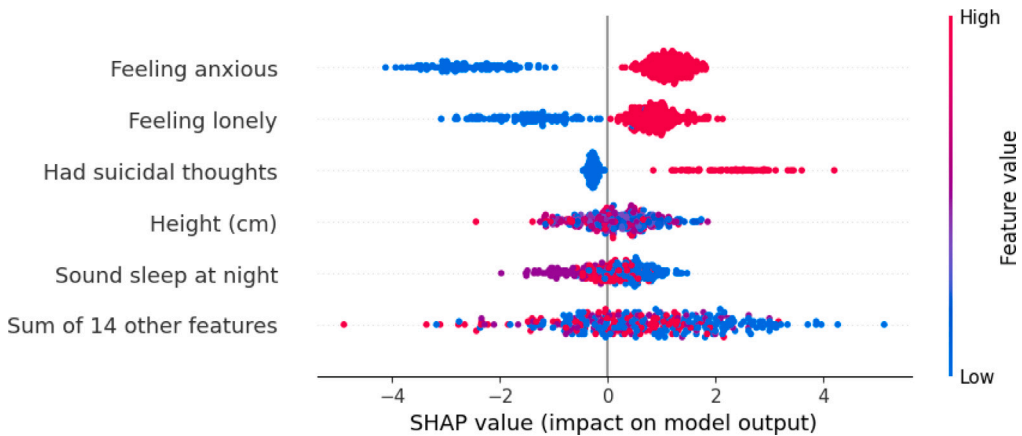


Fig. 13. SHAP beeswarm plot showing top 5 important features.

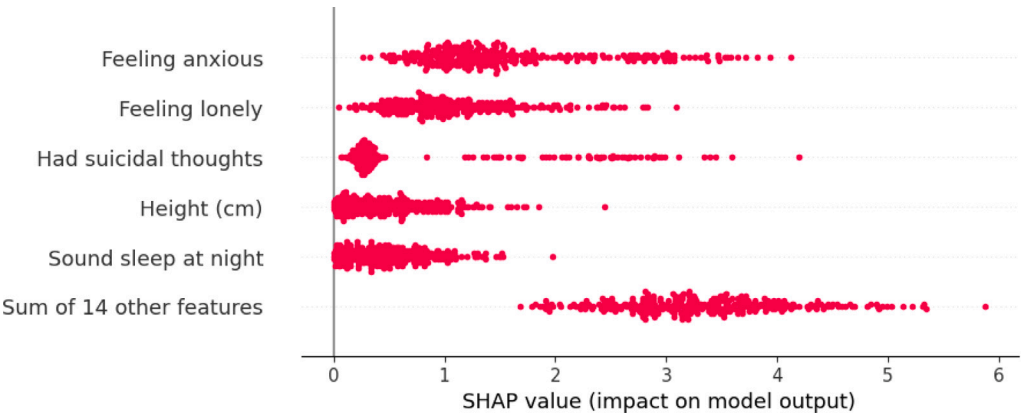


Fig. 14. SHAP beeswarm plot with absolute values showing top 5 important features.

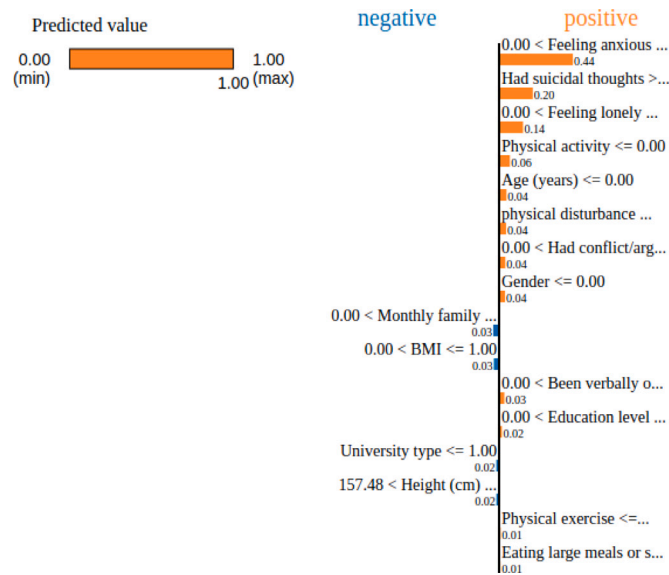


Fig. 15. LIME interpretation of a positive depression prediction case.

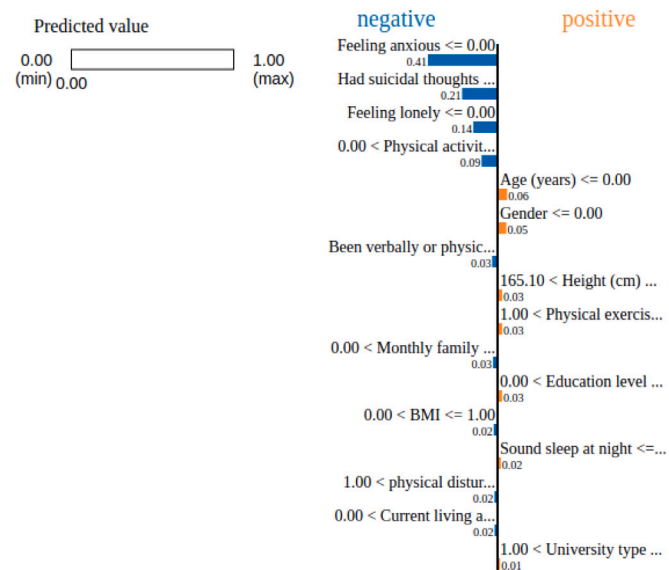


Fig. 16. LIME interpretation of a negative depression prediction case.

how well the models adapt over time and whether continuous learning approaches can improve long-term performance.

Beyond these technical and dataset-related limitations, cultural and linguistic factors may also pose challenges in adapting NLP models for this specific context. Variations in language use, mental health stigma, and the way depressive symptoms are expressed could influence model performance. Without culturally adapted approaches, models trained on one demographic may not generalize well to another. Exploring domain-specific pre-training, culturally aware NLP techniques, or fine-tuning on region-specific datasets to enhance model robustness across diverse populations can be a promising area of research.

6. Conclusion

This study highlights the substantial potential of machine learning and deep learning algorithms in predicting depression and provides valuable insights into the factors influencing mental health among university students in Bangladesh, considering both male and female

populations. By examining demographic, social, and academic variables, seven machine learning models, two deep learning models, and three transformer-based language models were employed to predict depression. Among these, Random Forest outperformed other models, achieving an accuracy of 91.1%. In terms of recall, language models demonstrated superior performance, with RoBERTa achieving a recall score of 98.6%.

Given the highly imbalanced nature of the dataset, resampling techniques were applied to address this challenge. To ensure the reliability and transparency of the predictions, we employed two distinct explainable AI methods — SHAP and LIME — to interpret the predictive outcomes. These methods revealed the most significant factors contributing to depression, providing actionable insights. Overall, this research emphasizes the critical role that AI can play in mental health assessments. By identifying key risk factors, it lays the foundation for targeted interventions aimed at improving the mental well-being and productivity of university students.

Future research can enhance depression detection by curating larger and more diverse datasets that encompass various demographic backgrounds. Training AI models on such datasets would likely improve predictive performance and generalizability. Since AI models are adaptable, fine-tuning them on datasets from different contexts can further refine their effectiveness. Additionally, leveraging generative large language models (LLMs) [40], such as GPT or LLaMA, presents another promising direction for identifying depression among students. Future work could also focus on deploying these models in real-world applications, such as counseling services or mental health platforms, to facilitate early detection and intervention. By incorporating these advancements, AI-driven depression detection could significantly contribute to improving mental health support systems.

CRedit authorship contribution statement

Gazi Hasan Al Masud: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Formal analysis, Data curation. **Rejaul Islam Shanto:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Ishmam Sakin:** Writing – original draft, Software, Methodology, Data curation. **Muhammad Rafsan Kabir:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Ethics statement

This study does not require approval from an ethics committee, as the dataset used is publicly available, anonymized, and curated with participant consent. Furthermore, our research does not involve any direct interaction with human subjects, ensuring there are no ethical concerns.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.array.2025.100375>.

Data availability

The dataset employed for this study is a publicly available dataset. It can be accessed through <https://data.mendeley.com/datasets/4v4bv3r8hv/1>.

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